SpendWise Project Documentation

Project Title

SpendWise – A Simple Backend Learning Project using Node.js, Express, MongoDB, and JWT

Team Details Team ID: SWTID-2026-1542

Team Leader: Levin C J

Team Members:

Saravanan K

Sathish G

Saravana M

Shyam V

Project Overview

SpendWise is a backend application designed to help beginner developers learn how to build secure REST APIs using Node.js, Express, MongoDB, and JWT authentication. The project focuses on implementing user authentication and basic CRUD operations in a clean and simple way so students can understand how backend systems work in real-world scenarios.

Key Objectives • Provide secure user authentication using JWT • Enable basic CRUD operations on user data (transactions/records) • Demonstrate MongoDB integration using Mongoose • Show how protected routes work using middleware • Maintain a simple and beginner-friendly backend architecture

Target Users

•Beginner backend developers

•Students learning Node.js and Express

•Freshers preparing for technical interviews

•Anyone who wants a simple project to understand JWT and MongoDB

•Learners who want hands-on practice with REST APIs

Project Instruction

SpendWise is a backend-only project built using Node.js and Express.

Quick Start

npm init

npm install

npm run dev

Steps to Create the Project

•Choose or create a directory for the backend project

•Open terminal and navigate to that directory

•Initialize Node.js project using npm init

•Install required packages: express, mongoose, jsonwebtoken, bcrypt, cors, dotenv

•Create folders: models, routes, controllers, middleware, config

•Start the server using npm run dev

•Test APIs using Postman or Thunder Client at http://localhost:5000

Pre-Requisites

Software Requirements

Node.js – JavaScript runtime for backend

MongoDB – NoSQL database for data storage

npm – Package manager

Git – Version control system

Code Editor – VS Code or similar

Knowledge Requirements

•JavaScript ES6+

•Node.js and Express.js

•MongoDB and Mongoose

•JWT authentication

•Basic REST API concepts

•Git basics

Thumbnail for Project

A simple backend project thumbnail representing:

Node.js + Express + MongoDB + JWT + CRUD APIs

Technical Architecture

Backend Architecture (Node.js + Express)

• Controllers: Handle business logic for authentication and CRUD

• Models: Mongoose schemas for User and Transaction

• Routes: API endpoints for authentication and CRUD operations

•Middleware: JWT verification and request protection

•Configuration: Database connection and environment variables

Database Architecture (MongoDB)

Users Collection

_id, name, email, password (hashed), createdAt

Transactions Collection

_id, userId (FK), title/description, amount/value, createdAt

Relationship

Many transactions belong to one user

Features

1. User Authentication – JWT-based registration and login

2. CRUD Operations – Create, Read, Update, Delete transactions

3. Protected Routes – Secure APIs using JWT middleware

4. MongoDB Integration – Store and fetch data using Mongoose

Roles and Responsibilities

Student / Developer

• Write backend code

• Create APIs

• Test APIs using Postman or Thunder Client

• Push code to GitHub

System (Backend)

•Authenticate users using JWT

•Handle CRUD operations

•Store and fetch data from MongoDB

•Protect private routes

Project Setup and Configuration

1. Install required tools

Node.js

MongoDB

2. Create project folders

server folders (models, routes, controllers, middleware, config)

3. Install backend packages

Express

Mongoose

Cors

Jsonwebtoken

Bcrypt

Dotenv

Backend Development

• Setup Express server

•Create server.js file

•Create .env file and define PORT and DB URL

•Configure middleware (cors, express.json)

•Create register and login routes

•Generate JWT token

•Protect routes using middleware

•Create authentication and transaction CRUD routes

•Implement user and transaction schemas

•Implement CRUD operations

•Add error handling

Database Development

Configure MongoDB

•Use MongoDB Atlas or local MongoDB

•Store connection URL in .env file

•Connect using Mongoose

Create Database Connection

•Create a config file for DB connection

•Use mongoose.connect()

•Handle success and error cases

Create Schema and Models

User Model

Name

Email

Password

Transaction Model

UserId

Title / Description

Amount / Value

Timestamps

Project Execution

1. Start MongoDB

2. Run backend server

3. Test Register API

4. Test Login API

5. Copy JWT token

6. Use token to access protected CRUD APIs

7. Perform Create, Read, Update, Delete operations

Output Screenshots

• Register API

• Login API

• Create Transaction

• Get Transaction Data

• Update Transaction

•Delete Transaction

GitHub Repository: https://github.com/bscsalevin-hub/SpendWise.git

Project Demo Video:

https://drive.google.com/file/d/1KuFfLM4iLzwB0ITSDHjY-sHifHMEyutr/view?usp=sharing

Conclusion

SpendWise demonstrates backend development concepts including REST API creation, JWT authentication, database integration with MongoDB, and secure route handling. This project helps beginners understand real-world backend application development.